PROJECT RESULTS

Physics-Aware ML for Bio-Oil Pyrolysis

An honest benchmark and a working Pyrolysis Advisor for higher yield and lower oxygen

TEAM 2

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OUTCOMES AT A GLANCE

Three things this project delivers

01

An honest benchmark

We showed that typical published accuracy collapses when a model is tested on a biomass it has never seen.

02

Physics-aware models

Final yield and O/C models built on kinetics-based features and physical constraints that behave like real pyrolysis.

03

A working Advisor

A tool that recommends temperature and heating rate, shows the yield/oxygen trade-off, and refuses unfamiliar feedstocks.

Data: Zhang et al. 2022 (O/C, 122 biomasses) and Pascarella et al. 2025 (yield, ~160 studies)

Random splits make models look far better than they are

0.66 → −0.18

Yield R^2, random split → unseen biomass (XGBoost)

0.63 → 0.23

O/C R^2, random split → unseen biomass (GPR)

Hidden near-duplicate rows in compiled data inflate scores further unless clustered first.

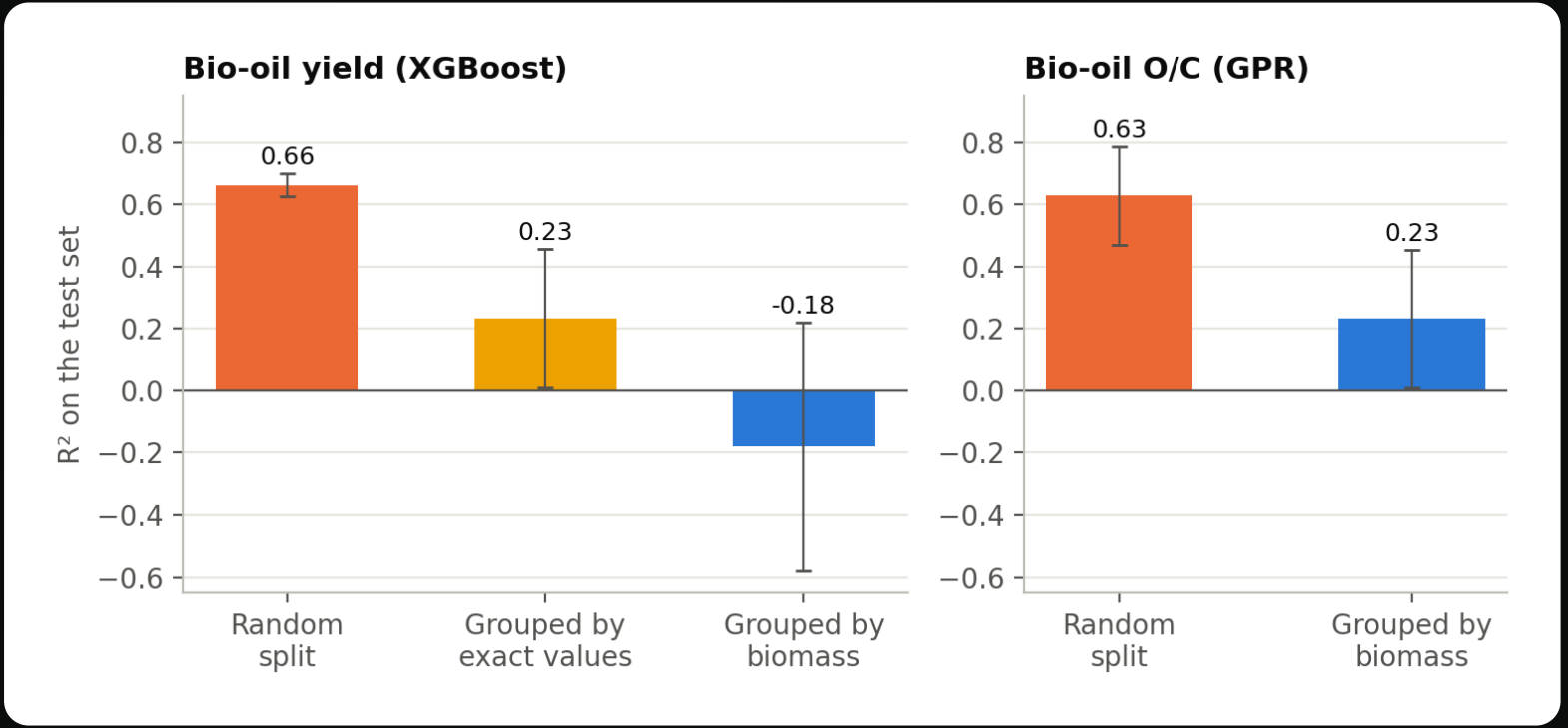
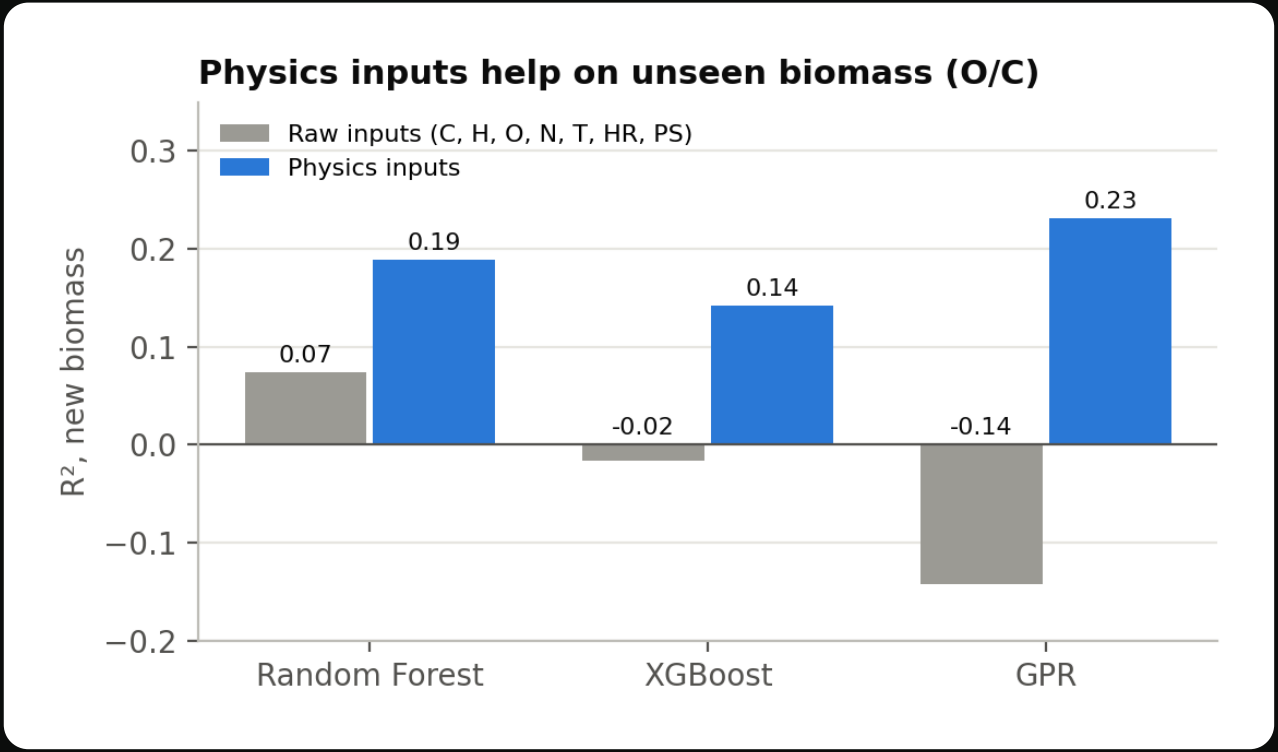


Figure: test R^2 under random vs. grouped-by-biomass testing

Physics-informed features help most on new biomass



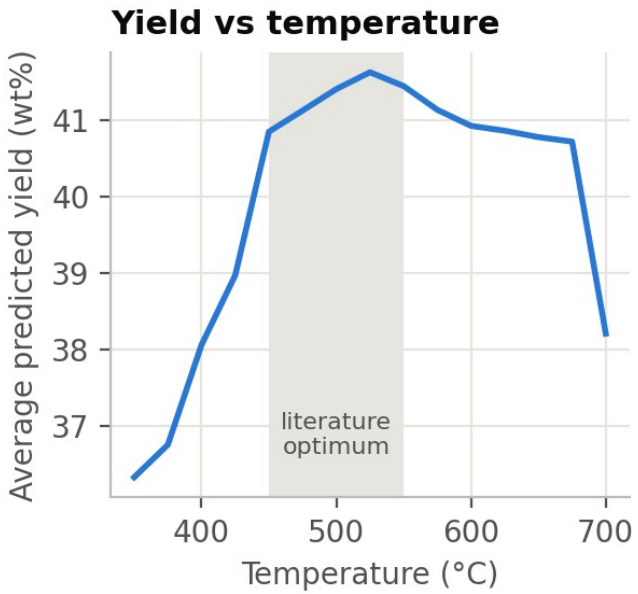
−0.14 → 0.23

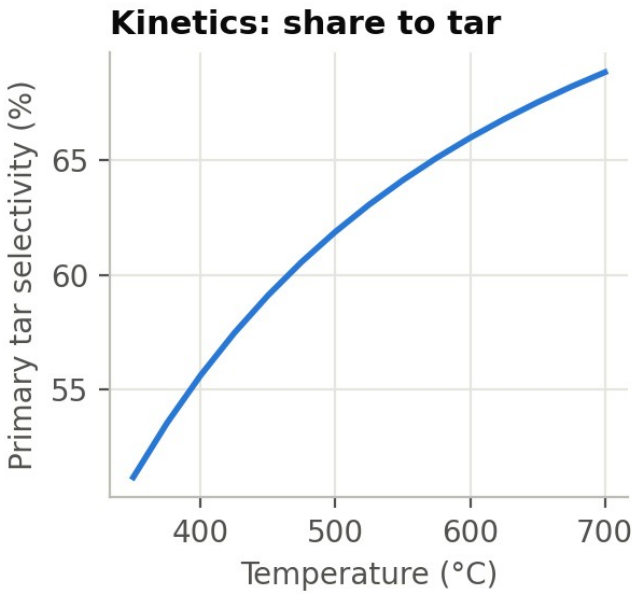
GPR O/C R^2 on unseen biomass: raw inputs → physics inputs

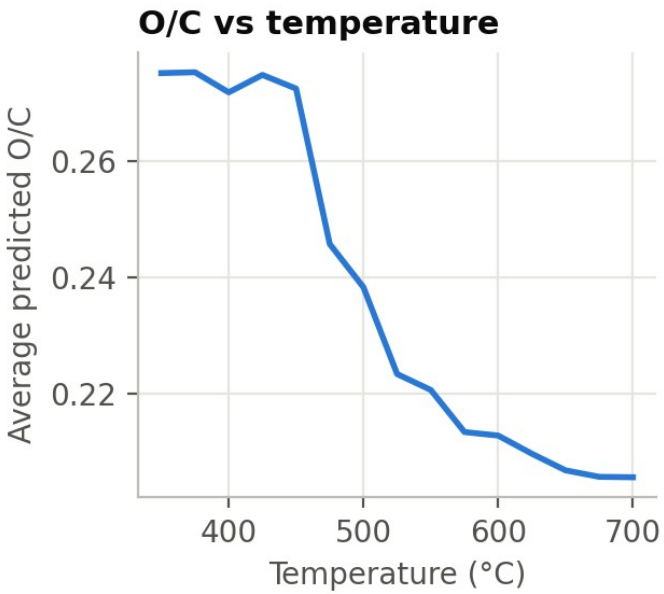
- Chan et al. (1985) Arrhenius kinetics: wood → gas / tar / char
- Kinetic tar share and pyrolysis number at each temperature
- Heating-regime and thermal-severity indices
- Molar O/C, H/C and effective H/C of the feedstock

RESULT 3 · PHYSICS CONSISTENCY

The final models behave like real pyrolysis







525°C

Yield peak (41.6 wt%) — inside the literature 450–550°C optimum

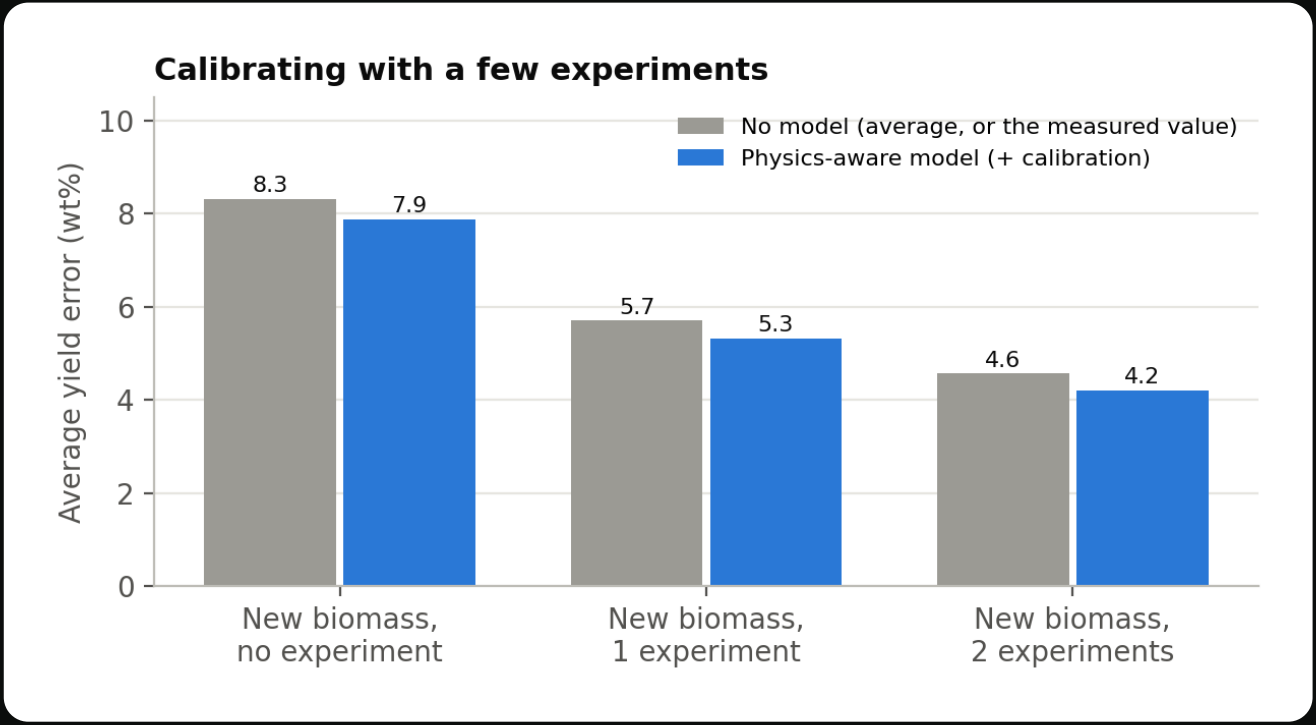
51% → 69%

Tar selectivity rises 350→700°C, explaining the yield drop from vapour cracking

0.28 → 0.21

O/C falls with temperature — stronger deoxygenation, never violated

One or two real experiments sharply cut the error



~8 → 5.3 → 4.2

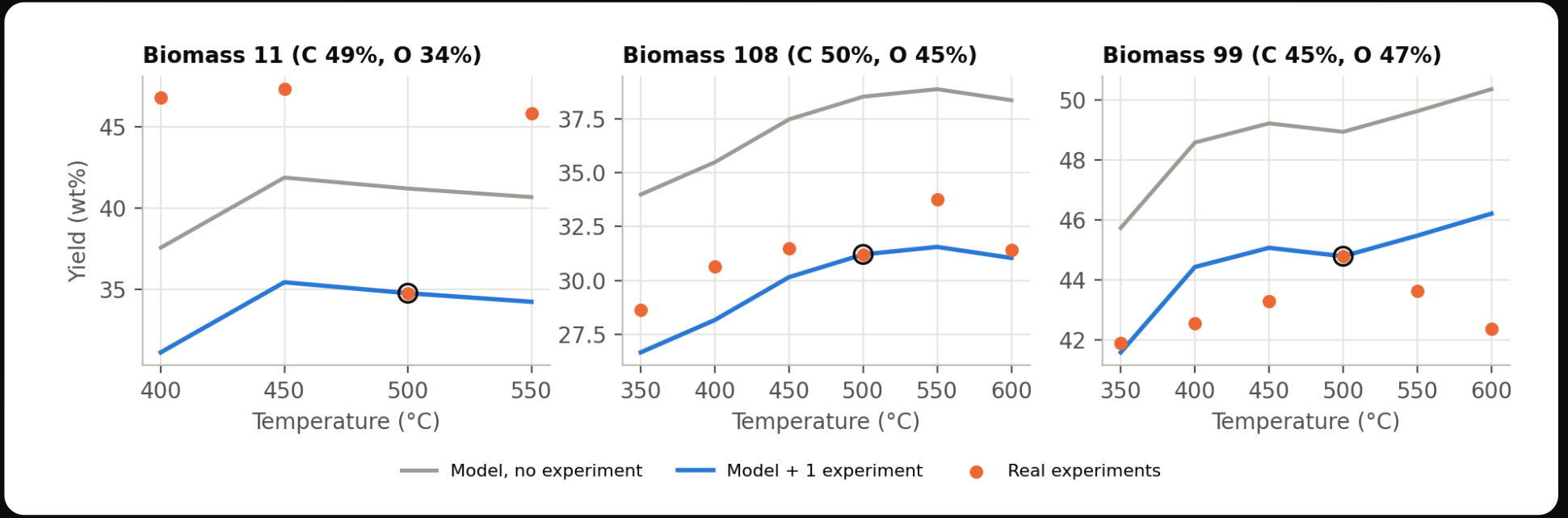
Average yield error (wt%) with 0, 1, 2 calibration experiments

±12.6 wt%

Honest 80% interval for a brand-new biomass with no experiments

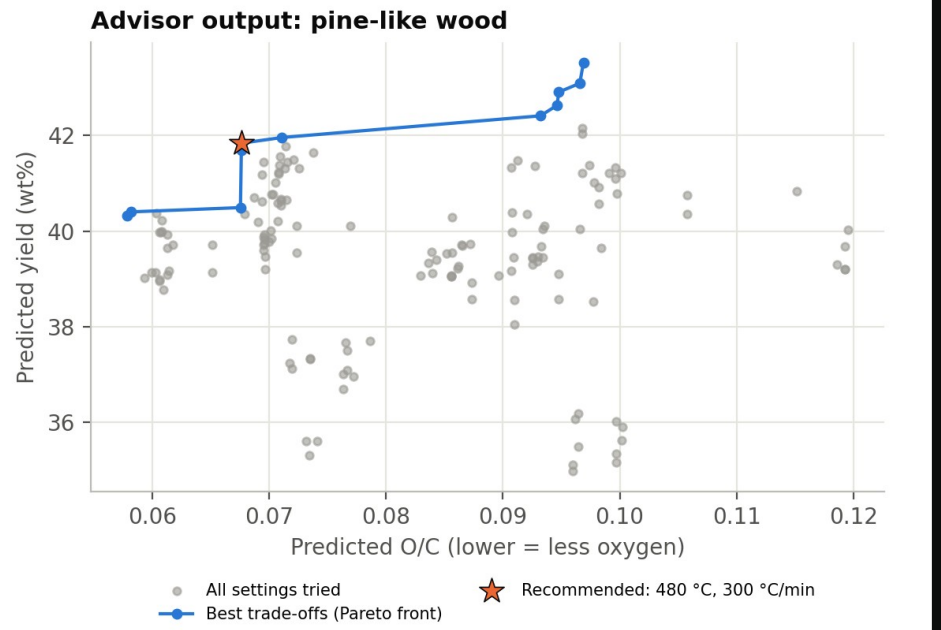
Calibrated model beats simply reusing the measured value (5.7, 4.6 wt%) because it adds the learned response to conditions.

Correct optimum temperature for 2 of 3 unseen biomasses



Biomass	Real best T	Model best T	Error, 0 exp.	Error, 1 exp.
11 (C 49%, O 34%)	450°C	450°C ✓	6.6 wt%	13.0 wt%
108 (C 50%, O 45%)	550°C	550°C ✓	5.6 wt%	1.7 wt%
99 (C 45%, O 47%)	500°C	600°C	6.0 wt%	1.9 wt%

Recommendations that show the trade-off — and know their limits

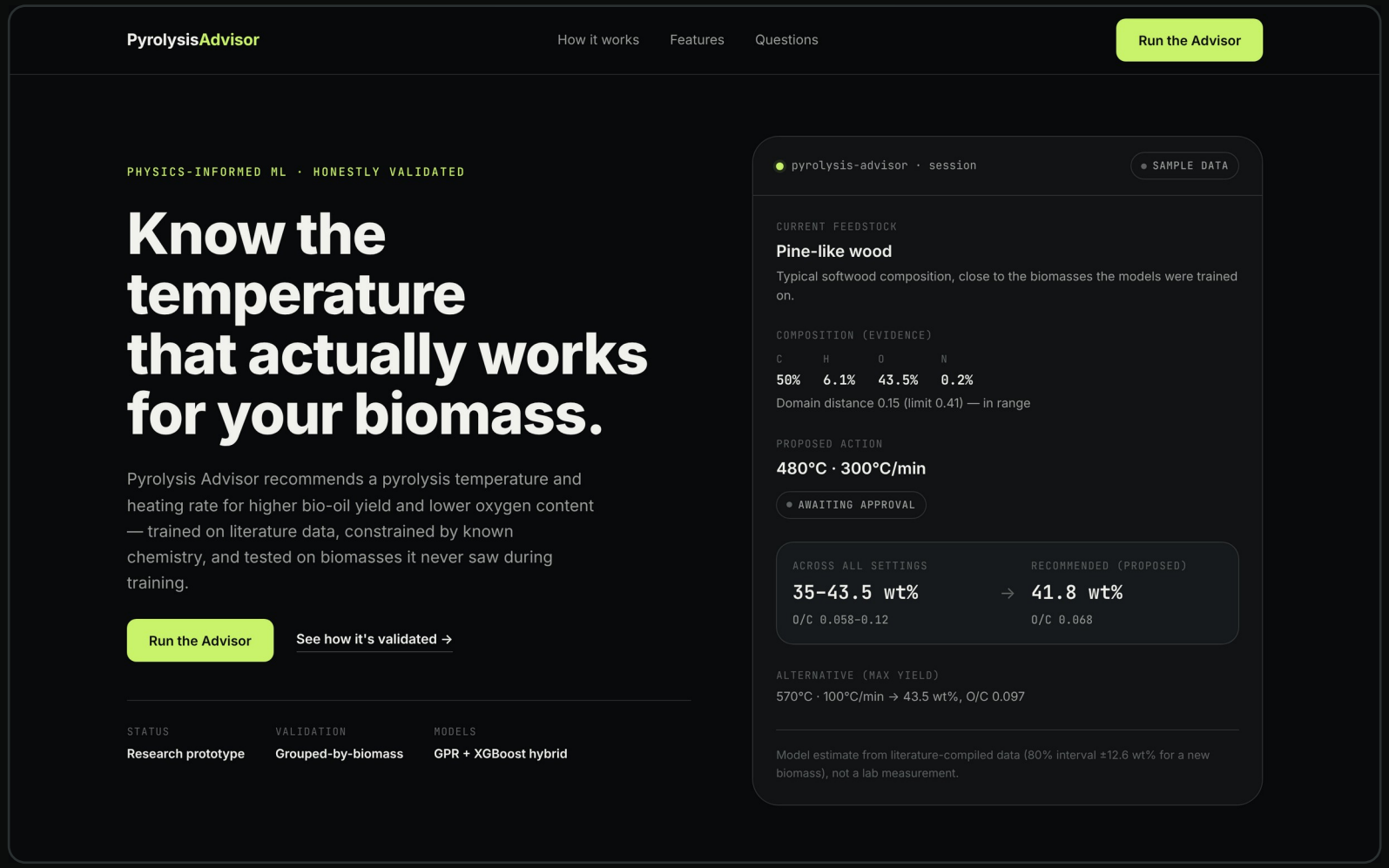


Feedstock	Status	Recommended	Max-yield alternative
Pine-like wood	In range	480°C, 300°C/min → 41.8 wt%, O/C 0.068	570°C → 43.5 wt%, O/C 0.097
Rice husk-like	In range	510°C, 10°C/min → 41.6 wt%, O/C 0.062	570°C → 44.7 wt%, O/C 0.113
Microalgae-like (N 8.5%)	Flagged	Withheld — distance 0.52 > 0.41	Needs 1–2 experiments
Coal-like (C 80%)	Flagged	Withheld — distance 1.20 > 0.41	Not biomass

For pine, the last 1.7 wt% of yield costs about 40% more oxygen in the oil.

WORKING MODEL · LIVE DEMO

The Advisor as a web app

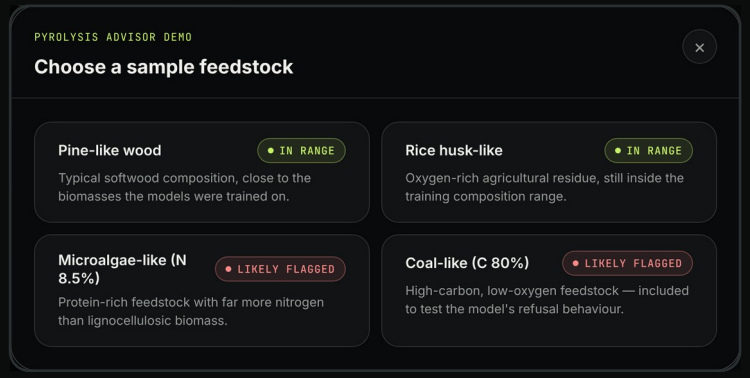


- React + TypeScript web app
- Pick a feedstock, review the recommended conditions, then accept or reject
- Every screen states it uses sample results from our validation

```
Run locally: advisor-frontend → npm run dev
```

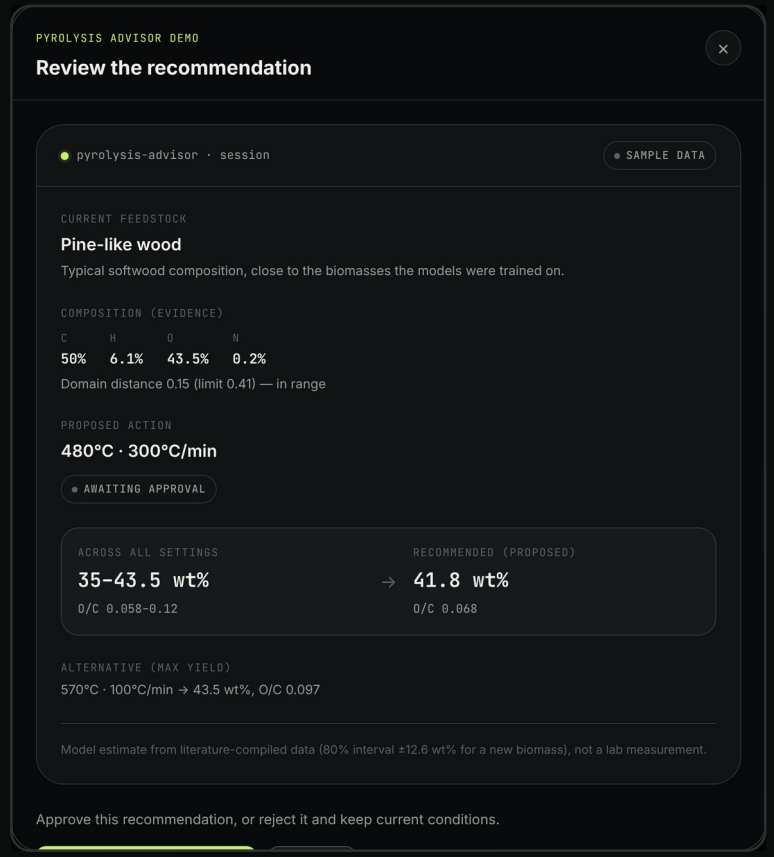
Choose → review → decide

1 Choose a sample feedstock

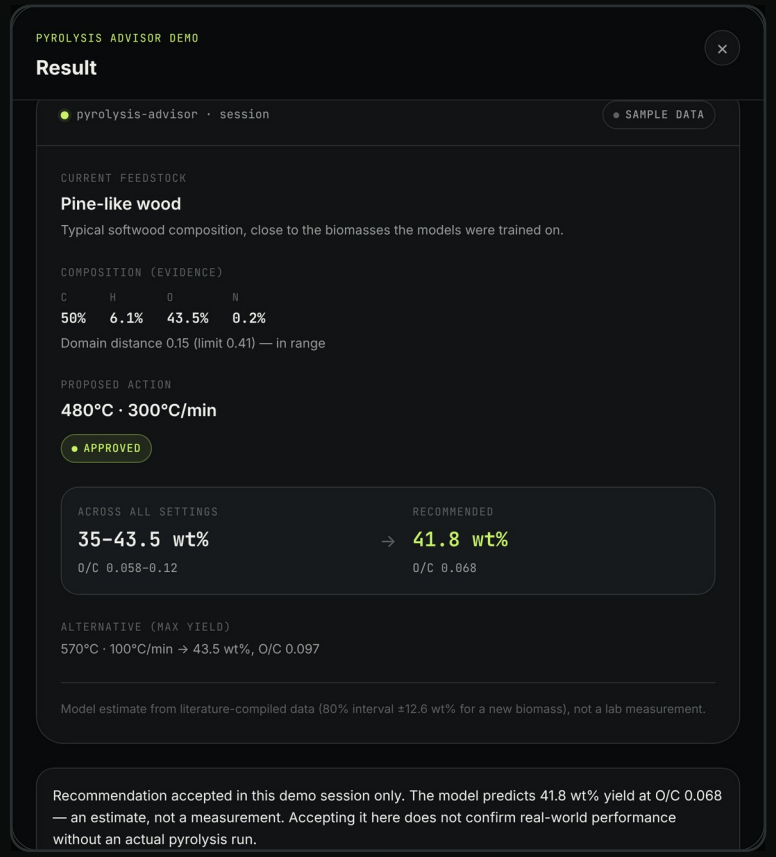


Two feedstocks are in range; two are flagged before any number is shown.

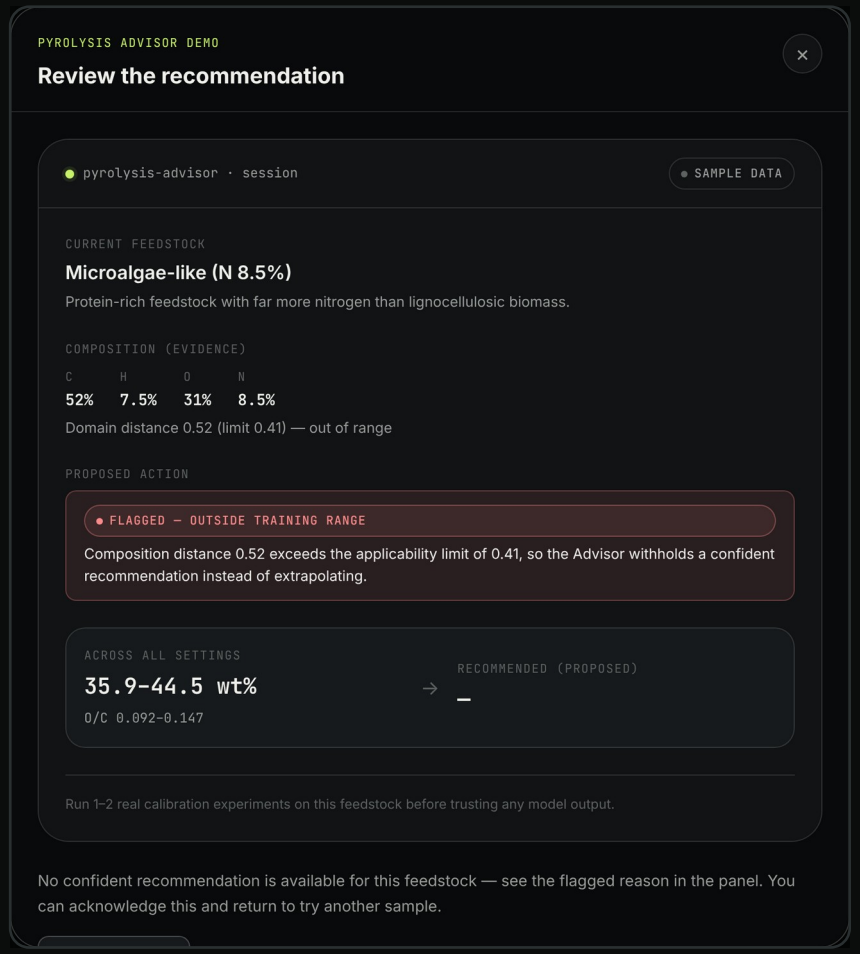
2 Review recommendation and range



3 Accept — result shown honestly



It refuses to guess on unfamiliar feedstock



0.52 > 0.41

Microalgae-like feedstock: composition distance exceeds the applicability limit

- No recommendation is shown, only the reason
- User is told to run 1–2 calibration experiments first
- Coal-like input (distance 1.20) is refused the same way

SUMMARY

Final honest scorecard

Metric (tested on unseen biomass)	Value	What it means
O/C R^2 (log scale), final blend	**0.27**	Real but modest skill; ~31% typical error
Yield R^2, final model	**0.07**	Yield level hard to transfer between labs
Yield shape R^2	**0.12**	Captures how yield responds to T and heating rate
Yield 80% interval, new biomass	**±12.6 wt%**	Uncertainty is reported, not hidden
Yield error after 1 / 2 experiments	**5.3 / 4.2 wt%**	Calibration makes it practically useful
Correct optimum T, case studies	**2 of 3**	With zero experiments
Out-of-range feedstocks refused	**2 of 2**	Microalgae and coal flagged

What we would do next

LIMITATIONS

- Literature data mixes labs; residence time rarely reported
- Yield and O/C come from two separate datasets
- Wide uncertainty for a brand-new biomass
- Web demo runs on pre-computed sample results

NEXT STEPS

- Leave-one-study-out testing to isolate lab effects
- Model organic (water-free) yield on 413 rows
- Collect consistent single-reactor data and retrain
- Connect the web app to the trained models via API

Thank you

Questions and live demo

github.com/umasuthanpalaniappan/BioOil-Automation